

Patricia L. Murphy

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Neuroscience Graduate Group
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EDUCATION

Doctor of Philosophy: Neuroscience, expected 2020
Certificate in University Teaching

University of Pennsylvania
Philadelphia, PA

Bachelor of Science: Biology, 2013
Highest honors, Minor: German, International Plan: French

Georgia Institute of Technology
Atlanta, GA

Erasmus Exchange: Life Sciences, 2012

Ecole Polytechnique Fédérale de Lausanne
Lausanne, CH

AWARDS

- 2017-2018 F31 National Research Service Award (National Institute of Neurological Disorders and Stroke)
- 2015-2016 T32 Developmental Biology Training Grant (UPenn)
- 2013-2015 T32 Systems and Integrative Biology Training Grant (UPenn)
- 2014 NSF Graduate Research Fellowship, Honorable Mention (also awarded in 2013)
- 2012 Georgia Tech President's Undergraduate Research Award (also awarded in 2010, 2011)
- 2012 Erasmus Exchange Scholarship (EPFL)
- 2012 Reginald S. Fleet Study Abroad Scholarship (Georgia Tech)
- 2010 Omicron Delta Kappa Leadership Award (Georgia Tech)
- 2009-2013 Georgia Tech President's Scholarship (top 60 incoming freshmen at Georgia Tech)
- 2009-2013 Penny and E. Roe IV Leadership Scholarship (top 10 incoming freshmen at Georgia Tech)

RESEARCH

Doctoral Thesis: **Role of Slit-Robo in target specific nerve regeneration**

University of Pennsylvania
Philadelphia, PA

Advisor: Michael Granato, PhD (Cell and Developmental Biology)

2014-2020

Skills: molecular cloning, confocal microscopy, live imaging, in situ hybridization, CRISPR, transgenic chimeras

Summary: To restore function after nerve injury, severed axons must reconnect to appropriate targets. Using **laser nerve transection** and **time-lapse imaging** in larval **zebrafish**, I discovered that the axon guidance receptor *robo2* is required to prevent and correct errors made by regenerating axons at spinal motor nerve branch points. I generated **transient and stable transgenics** and **CRISPR mutants** to place *robo2* in a molecular axon regeneration pathway with its canonical ligand Slit and the ECM component Type-IV Collage (*col4a5*). *Slit1a* and *col4a5* are expressed by glia at the nerve branch point during regeneration. My data suggest that these genes act in a molecular pathway with *robo2* to direct regrowing motor axons to the correct nerve branch through axon-glia interactions at the nerve branch point. Understanding how regenerating axons navigate nerve branch points is very important clinically, especially since therapies aimed at increasing axon regrowth often result in increased misdirection of regenerating axons.

Pre-doctoral Rotation: **Cell culture and biomaterial approaches to transplant pre-formed neural networks into the brain**

University of Pennsylvania
Philadelphia, PA

Advisor: D. Kacy Cullen, PhD (Neurosurgery)

Spring 2014

Skills: 2D and 3D embryonic and adult rat primary cortical neuron and astrocyte cell culture, biomaterial tissue culture scaffold fabrication and testing, immunocytochemistry, confocal microscopy

Summary: To repair damaged neural circuits in the brain, the Cullen Lab has developed **micro-tissue engineered neural networks** (micro-TENNs) composed of primary neurons suspended in artificial extracellular matrix (ECM) and encased in a hydrogel tube. Two major challenges in micro-TENN fabrication were to obtain reproducible cellular architecture within hydrogel tube and to develop minimally invasive methods to deliver preformed micro-TENNs into the brain. By optimizing the ECM composition and method of **cell delivery**, I reliably obtained the desired cellular architecture. I also developed methods to coat micro-TENNs in carboxymethyl cellulose (CMC). Following mild dehydration, CMC-coated micro-TENNs were stiff enough to withstand forces necessary for **needle-less delivery into the rodent brain**. My work on this project was **published in Journal of Neural Engineering** (Harris, et al., 2016; Figures 3, 5, 7). I was also the first person in the lab to seed astrocytes in hydrogel tubes and to notice that they adopt radial glia-like morphology in these conditions. This observation led to a fruitful and ongoing area of study in the lab.

Undergraduate Thesis: **Identifying Neuronal Receptors for CSPGs**

Advisor: Ravi Bellamkonda (Biomedical Engineering)

2010-2013

Georgia Institute of Technology

Atlanta, GA

Skills: membrane-protein purification, western blot, rat primary cortical neuron harvesting and cell culture, immunofluorescent-fluorescent in situ hybridization (IF-FISH), confocal microscopy

Summary: After injury to the central nervous system (CNS), chondroitin sulfate proteoglycans (CSPGs) are upregulated and known to inhibit axon outgrowth. Until 2009, there were no known neuronal receptors for CSPGs. The central aim of my thesis was to identify additional neuronal receptors for CSPGs and downstream mechanisms of their inhibitory effects on neurite outgrowth. Using IF-FISH, I demonstrated that RhoA is locally translated and activated in axons growing on a CSPG substrate. This suggests that CSPGs mediate axon stalling and retraction through local regulation of cytoskeletal dynamics. We performed tandem mass spectrometry on cortical membrane proteins that bound the CSPG aggrecan and used peptide fingerprinting to identify the Leukocyte Antigen Related-phosphatase (LAR) as a promising CSPG receptor candidate. We validated that LAR binds CSPGs using surface plasmon resonance (SPR). Jerry Silver's lab separately identified and published LAR as a CSPG receptor that activates RhoA in late 2011.

Institute on Neuroscience Fellowship: **C3 transferase inhibition of RhoA: lentiviral vector mediated strategy for axon regeneration**

Emory University
Atlanta, GA

Advisor: Robert E. Gross, MD PhD (Neurosurgery)

2008-2009

Skills: molecular cloning, cell culture (HEK293T, PC12), transfection, western blot

Summary: To improve axon regeneration after central nervous system injury, the Gross Lab designed viral vectors encoding C3 transferase, a bacterial enzyme which inhibits RhoA. The lab had previously shown that C3 transferase inhibition of RhoA resulted in neurite outgrowth in the presence of chondroitin sulfate proteoglycans (CSPGs), which are known to inhibit axon regeneration. My specific role was to clone a self-cleaving peptide into the lab's viral vector to separate C3 transferase from a fluorescent tag. Compared to the direct fusion construct, I found that cells transfected with cleaved C3 transferase extended more neurite-like projections. I wrote a manuscript reporting my results which was selected for publication in National Consortium for Secondary Schools in STEM journal. In addition, I won a spot as a delegate to the National Junior Science and Humanities Symposium for an oral presentation of this work.

K-12 TEACHING

Elementary School Neuroscience Outreach Program | 2013-2018

Independence Charter School

Founder and Lead instructor

Philadelphia, PA

- Led development of neuroscience curriculum for grades K-4 in coordination with primary school science teacher.
- Implemented curriculum over five consecutive years with grade-specific lessons team-taught by graduate student volunteers.
- Reached >800 students with >200 completing full K-4 sequence.
- Program ongoing under new leadership. Selected programming available online: <https://www.upenn.org/grades-k8>.

PepsiCo Innovation Challenge: Leadership in STEM | 2017

New York Academy of Sciences

Lead Instructor

New York, NY

- Independently developed intensive half-day workshop on leadership and communication in STEM.
- Led team of 3 facilitators (10 students, grades 10-12).

STEM Scholars Neuroscience Workshop | 2015-2017

The Franklin Institute Science Museum

Instructor

Philadelphia, PA

- Co-developed full-time week-long summer workshop with introductory neuroscience lectures and hands-on activities.
- Co-taught with one other neuroscience graduate student (20 students, grades 9-12).

Upward Bound Neuroscience Elective | 2014-2017

University of Pennsylvania

Instructor

Philadelphia, PA

- Co-developed summer course based on original textbook developed in collaboration with other Penn neuroscience graduate students (*Neuroscience: A Primer*, ISBN: 1329062450).
- Co-taught lessons on Neuroscience in the News, Stress, Neurotransmission (20 students, grades 9-12).

Pre-College Leadership Institute | 2015-2016

New York Academy of Sciences

Instructor

New York, NY

- Co-developed full-time week-long intensive workshop on STEM communication, leadership and research methods.
- Guided students in original research proposals. Co-facilitated in team of 4 (60-80 students, grades 9-12).

Penn Summer Research Academies | 2015-2016

University of Pennsylvania

Instructor

Philadelphia, PA

- Designed three-week summer seminars on primary scientific literature in neural tissue engineering (2016) and motor nerve regeneration (2015). Independently taught 9-10 students, grades 9-11.

UNIVERSITY TEACHING

Center for Teaching and Learning: TA Training | 2018

University of Pennsylvania
Philadelphia, PA

Teaching Assistant Trainer

- Co-led university-wide, week-long training for new graduate teaching assistants.
- Designed, taught workshops on Grading in the Sciences (25 students), Leading Discussions in the Sciences (18 students).
- Facilitated individual teaching demonstrations and group discussions of pedagogical approaches (10 students).

Introduction to Brain and Behavior (BBB109) | Spring 2017

University of Pennsylvania
Philadelphia, PA

Head Teaching Assistant

- Mentored 6 graduate, 2 undergraduate Teaching Assistants in pedagogical approaches by observing, discussing recitations.
- Managed logistics of 2 weekly lectures, 8 weekly recitations and 4 labs for 168 undergraduate students.
- Supervised and mentored TAs writing, grading 3 exams.

Molecular and Cellular Neuroscience (BBB251) | Fall 2015

University of Pennsylvania
Philadelphia, PA

Teaching Assistant

- Taught weekly recitations, including 4 electrophysiology labs to 22 undergraduate students.
- Wrote and graded 3 exams in team of 8 teaching assistants (147 students).

Cell Biology and Biochemistry (BIOM600) | Fall 2015-2016

University of Pennsylvania
Philadelphia, PA

Tutor

- Led weekly exam review for 5-8 graduate students.

Invited Lectures

2018	Axon guidance in development PYSCH205: Neuroscience Foundations (15 undergraduate students)	St. Joseph's University Philadelphia, PA
2017	Neurogenesis and axon guidance in development BBB109: Introduction to Brain and Behavior (168 undergraduate students)	University of Pennsylvania Philadelphia, PA
2015	Skype discussion: Isaacman-Beck, et al. 2015, <i>Neuron</i> BIOL148L KS: Neuroscience I: Cell and Molecular (30 undergraduate students)	Scripps College Claremont, CA

INDIVIDUAL TEACHING AND MENTORING

Designed rotation and science fair projects, provided technical help and support

Fall 2019	Sophia Villiere, neuroscience graduate student, Granato Lab Rotation, UPenn
Fall 2017	Leah Middleton, neuroscience graduate student, Granato Lab Rotation, UPenn
Summer 2017	Daniel Morales, undergraduate at Boston University, Granato Lab Internship, UPenn
Summer 2016	Chelsea Alexander, high school senior, STEM Outreach Internship, UPenn
2015-2016	Anna Roberts, high school junior, Granato Lab Internship, UPenn
2015-2016	Rupa Pendyala, high school sophomore, MIT THINK semi-finalist 2016
2014	Dayo Adewole, bioengineering masters student, Cullen Lab Rotation, UPenn

STEM OUTREACH

K-12 Outreach (grade)

2020	Bioeyes, Cook-Wissahickon School, presented original research (8)
2019	Girls Advancing in STEM Conference, UPenn, presented original research (10-12)
2019	DrosoPhila, Northeast High School, taught lesson on axon guidance in flies (10)
2018	DrosoPhila, Benjamin Rush High School, taught lesson on flies as model organisms (10)
2018-2019	Granato Lab visits for high schoolers: organized live imaging demonstrations (12)
2018	Bioeyes, Masterman Demonstration School, presented original research (11)
2017	Bioeyes, Central Philadelphia High School, presented original research (11)
2015-2016	UPenn Kids Judge! organized and exhibited Left Brain vs. Right Brain demo (4-6)
2015	Granato Lab visit for high schoolers: developed and taught hands-on genetics lesson (9)
2015	"NMJ Game," John Hancock Demonstration School (5 th grade), developed and facilitated (5)
2015	"Using art to study network neuroscience," Samuel B. Huey Elementary School (6)
2015	Penn Alexander School Science Fair, judge (4-5)
2014-2016	Upward Bound Math and Science, UPenn, Research Friday's mentor (9-12)
2013	iPraxis middle school science fair program, mentor (7)
2010-2013	Rockdale Magnet School Regional Science Fair, judge (9-12)
2010-2011	Georgia Tech High School Math Competition, proctor (9-12)
2006-2010	Rockdale Magnet School Math Competition, proctor (9-12)

Public Outreach

- 2016 Philadelphia SfN Brain Awareness Week, Franklin Institute Museum, organizer and exhibitor
- 2015-2018 Clark Park Science Discovery Day, auditory illusions exhibitor
- 2015-2017 Philadelphia Science Festival, memory and EEG exhibitor
- 2015 Science After Dark, Franklin Institute Science Museum, memory and EEG exhibitor
- 2014-2016 Neuroscience Public Lecture Series, UPenn, organizer (TEDx-style events on TBI, addiction)

LEADERSHIP AND SERVICE

Brains in Briefs upenn.glia.com/briefs | 2018-2019

University of Pennsylvania
Philadelphia, PA

Co-Director

- Rescued and formalized blog about Penn Neuroscience graduate students' recent publications.
- Led 28 writers and 8-person leadership team in developing system for frequent and sustainable publication.
- Organized monthly writing workshops for peer-to-peer feedback on science writing for general audience.

Graduate Student-Led Initiatives and Activities Committee | 2015-2018

University of Pennsylvania
Philadelphia, PA

Executive Board

- Served three annual terms as elected **Outreach Chair, Co-Director, Treasurer.**
- Allocated and oversaw management of \$20,000 budget by four committee chairs (Outreach, Community Building, Professional Development and Public Relations) and 9 Outreach initiative leaders.
- Served as liaison between Neuroscience graduate students and faculty.

Biomedical Graduate Studies | 2015-2018

University of Pennsylvania
Philadelphia, PA

Institutional Committees

- Improv for Scientists mini-course planning committee (co-chair, 2018)
- Diversity in STEM (Penn-wide RCR event) planning committee (member, 2017)
- Moorman-Simon Seminar on Undergraduate STEM Education (panelist, 2016)
- Student Invited Speaker Selection Committee (co-chair, 2016)
- Neuroscience Student Retreat Planning Committee (co-chair, 2015)

Stamps Scholars National Convention | 2011

University of Miami
Miami, FL

Chair of Speaker Selection

- Co-designed and managed three-day inaugural conference for Stamps Leadership Scholars.
- Attended by >100 students, >50 academic professionals. Budget >\$200,000.

TEAM Buzz: annual day of service | 2010-2011

Georgia Institute of Technology
Atlanta, GA

Director of International Projects

- Expanded Georgia Tech's annual day of community service (est. 1998) to include ESL and STEM instruction at secondary schools in France, Singapore, and China (>50 volunteers/year).

PUBLICATIONS

Journal Articles

Harris, JP, Struzyna, LA, **Murphy PL**, Adewole, DO, Kuo, E, Cullen DK. (2016). Advanced biomaterial strategies to transplant pre-formed micro-tissue engineered neural networks into the brain. *Journal of Neural Engineering*, 13: 016019.

PRESENTATIONS

International Conferences

Oral Presentations

(Selected) Slit-Robo functions through Type-IV Collagen to promote target-specific axon regeneration. *The Allied Genetics Conference*, National Harbor, MD, April 23, 2020.

Roundabout2 and Collagen-IV function in a common molecular pathway to guide regenerating axons. *International Zebrafish Conference*, Madison, WI, June 21, 2018.

Roundabout2 promotes target specific peripheral nerve regeneration in vivo. *The Allied Genetics Conference*, Orlando, FL, July 16, 2016.

Roundabout2 and exotosin-like 3 promote target specific peripheral nerve regeneration in vivo. *Keystone Symposium: Axons from Biology to Pathology*, Santa Fe, NM, January 25, 2016.

International Conferences, cont.

Poster Presentations

Murphy PL, Granato MG. Roundabout2 and Collagen-IV function in a common molecular pathway to guide regenerating axons. *Molecular Mechanisms of Neuronal Connectivity* (Abstract No.122), Cold Spring Harbor, NY, September 24-29, 2018. ****Poster award**

Murphy PL, Isaacman-Beck J, Granato MG. Roundabout2 promotes target specific peripheral nerve regeneration in vivo. *Axon Guidance, Synapse Formation, and Regeneration*, (Abstract No.112), Cold Spring Harbor, NY, September 20-24, 2016.

Reiner DJ, Christison-Lagay K, Folweiler K, Healey M, Kahn JB, **Murphy PL**, Parthasarathi T, Perron IJ, Taylor MM. Graduate-led outreach initiatives: improving neuroscience literacy in children and adults. *Society for Neuroscience*, Program No. 26-14323. San Diego, CA: November 12-16, 2016.

Murphy PL, Isaacman-Beck J, Syrett C, Granato MG. Roundabout2 and exotosin-like 3 promote target specific peripheral nerve regeneration in vivo. *Keystone Symposium: Axons from Biology to Pathology*, Abstr J4-2002, Santa Fe, NM, Jan 24-26, 2016.

Harris, JP, Struzyna, LA, **Murphy PL**, Cullen DK (2014) Advanced biomaterial strategies for micro-tissue engineered neural networks to restore brain circuits. *National Neurotrauma Society Symposium*, Abstr 10311, San Fransisco, CA, June 29 – July 3, 2014.

Local Conferences

Oral Presentations

Roundabout2 and Collagen-IV function in a common molecular pathway to guide regenerating axons. *Mid-Atlantic Zebrafish Meeting*, Philadelphia, PA, December 7, 2018.

C3 transferase inhibition of RhoA: lentiviral vector mediated axon regeneration therapy. *Georgia Junior Science and Humanities Symposium*, Athens, GA, February 2-4, 2009. ****3rd place overall**

Poster Presentations

Murphy PL, Granato MG. Roundabout2 and Collagen-IV function in a common molecular pathway to guide regenerating axons. *Mahoney Institute for Neuroscience Symposium*, Philadelphia, PA, April, 2019.

Murphy PL, Granato MG. Roundabout2 and Collagen-IV function in a common molecular pathway to guide regenerating axons. *Cell and Developmental Biology Retreat*, Galloway, NJ, May 3-4, 2018.

****Poster award**

Murphy PL, Isaacman-Beck J, Granato MG. Roundabout2 promotes target specific peripheral nerve regeneration in vivo. *Mid-Atlantic Zebrafish Meeting*, Bethesda, MD, December 9, 2016.

****Poster Blitz Presentation**

Murphy PL, Isaacman-Beck J, Syrett C, Granato MG. Roundabout2 and exotosin-like 3 promote target specific peripheral nerve regeneration in vivo. *Mahoney Institute for Neurosciences Symposium*, Philadelphia, PA, April 27, 2016. ****Poster award**

PROFESSIONAL SOCIETIES

International Zebrafish Society, 2018-Present

New York Academy of Sciences, 2015-2019

Genetics Society of America, 2016-Present